

NO-LOAD And OVERLOAD PROTECTOR For AC Motors

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There are occasions when the water pump runs idle when there is no water in the sump/underground tank or the treadmill machine is left on, unintentionally. In such cases, motors of these electrical machines run unnecessarily. Running these for a long time may lead to burn-out. Such situations can be addressed by using this no-load and overload protector circuit. It can switch off the appliance in case of overload. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the no-load and overload protector for the AC motorised circuit is shown in Fig. 2. It is built around NE555 timer (IC1), decade counter 4017 (IC2), quad comparator LM339 (IC3), quad NOR gate CD4001 (IC4), current transformer (CT1), transistors BC547 (T1, T3, T4) and SL100 (T2), and a few other components. The circuit works off a 12V DC supply. You can use a 12V, 1A SMPS unit at connector CON1.

Clock signal is provided to decade counter 4017 (IC2) from the astable multivibrator wired with timer NE555 (IC1) through NOR gate N4. Other input of N4 is fed from Q4 output of IC2. If Q4 of IC2 is low, clock pulse enters counter IC2.

In a normal situation, Q4 of IC2 will be at low state, except for no-load condition. Q0 output of IC2 will be high if reset pin 15 of IC2 is in high state. When reset pin 15 is in low state, it allows the counter to count, so each output (Q0 through Q3) will change the state from low to high, sequentially. All four outputs (Q0 to Q3) are logically ORed and used to drive T2 and energise relay RL1.

If Q4 of IC2 becomes high, NOR gate N4 will not allow the clock pulse to enter the counter. So, T2 will be off

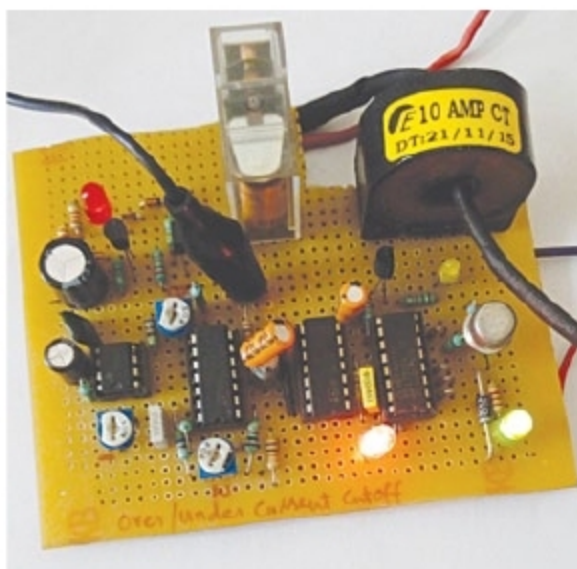


Fig. 1: Author's prototype

TRUTH TABLE OF NOR GATE		
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

and RL1 will be de-energised. This condition will continue till the next power supply restarts.

The amount of load current to the motorised electrical gadget is sensed by the coil of 50Hz, 10A current transformer (CT1) in the form of AC voltage. Sampled AC voltage in CT coil is proportional to the load current of the electrical gadget. This 50Hz AC voltage across the coil is rectified and filtered to get corresponding DC sample voltage. DC sample voltage is compared with reference voltages derived through two resistive networks in two comparators, A1 and A3, of LM339 (IC3).

In normal load condition, voltage from potmeter VR2 (which is used for checking no-load condition) should be set marginally less than the sample DC voltage. This provides high output at pin 2. If load current reduces, refer-

ence voltage from VR2 goes higher than the sample voltage, and output of A1 at pin 2 becomes low.

In normal load condition, reference voltage from VR3 is set adequately higher than the sample DC voltage. This gives low output at pin 14 of A3. If an overload condition arises, sample voltage goes higher than reference DC voltage, and output at pin 14 of A3 becomes high.

Gate N2 of IC4 is wired as a NOT (inverter) gate for making an initial reset pulse circuit with RC network formed by R12 and C6. This reset circuit is essential to reset counter IC2 on initial power on, so that output Q0 of IC2 becomes high and switches on T2. This, in turn, energises relay RL1 to provide AC power to the load.

DC sample voltage from the rectifier is available at cathode of D1. It is used for comparison and logical operation of gates. Normally, output of A3 at pin 14 is low, and capacitor C7 is in discharged state. Zener diode ZD1 will not conduct, and SCR1 will be cut-off. T4 switches on as it gets base bias through R16, LED3, D7, D8 and R20.

T3 goes off as its base is grounded through T4. Off state of T3 will not pull down the base bias voltage of T2. T2 is controlled by the output levels from Q0 to Q3 of IC2. The table shows the truth of NOR gate.

Operation of the circuit on different load conditions

Normal-load current condition. In this condition, DC sample voltage from the rectifier at pins 5 and 9 of IC3 are more than its reference voltage at pin 4, and less than its reference voltage at pin 8. Since load current differs for various electrical gadgets, for example, different horse power rated motors of water pumps, reference voltages at pins 4 and 8 have to be set using VR2 and VR3